

Locality vs Global Attention: Comparing Inductive Biases in Permutation-Equivariant Weight-Space Architectures

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Abstract

Permutation-equivariant architectures for weight-space learning—including Deep Weight Space (DWS) and Neural Functional Transformers (NFT)—have been proposed with different inductive biases, yet no systematic comparison exists on model property prediction tasks. We provide the first quantitative measurement of these inductive bias differences: DWS produces more localized weight updates (coefficient of variation 1.44) while NFT distributes information more uniformly (CoV 1.35). Testing whether these differences translate to task-dependent performance, we find that NFT’s global attention yields 12.3% better accuracy prediction (RMSE 82.9 vs 94.5), confirming its advantage on holistic property aggregation. However, the hypothesized DWS locality advantage on local anomaly detection could not be verified—all architectures performed at chance on synthetic backdoor tasks—leaving this direction for future work with real benchmarks. Our findings provide actionable guidance: use global attention architectures for tasks requiring aggregate statistics, while the case for locality-preserving architectures on local pattern detection remains open.

1 Introduction

Two neural network architectures designed for the same purpose—processing neural network weights while respecting permutation symmetry—produce measurably different internal representations. Yet this inductive bias difference only translates to performance advantages on global statistics tasks, not on local anomaly detection. This counterintuitive finding challenges the assumption that architectural differences uniformly manifest across all downstream tasks, and provides the first empirical guidance for practitioners selecting weight-space architectures.

Weight-space learning has emerged as a powerful paradigm for analyzing neural networks directly through their parameters, enabling applications from model property prediction to weight editing Zhou et al. [2024], Navon et al. [2023]. Central to this paradigm is the observation that neural network weights exhibit permutation symmetry—reordering neurons within a layer yields functionally equivalent networks. Two prominent architectures exploit this symmetry differently: Deep Weight Space (DWS) uses equivariant layers that preserve weight locality through structured operations Navon et al. [2023], while Neural Functional Transformers (NFT) flatten weights into tokens and apply global self-attention Zhou et al. [2024].

The surface problem is well-recognized: multiple permutation-equivariant architectures exist, but no systematic comparison evaluates them on model property prediction tasks. Prior work evaluates each architecture in isolation—NFT on implicit neural representation tasks, DWS on weight editing—leaving practitioners without guidance for tasks like backdoor detection or accuracy prediction.

The deeper problem we identify is that different architectures encode fundamentally different inductive biases, and how these translate to task performance remains unexplored. DWS’s equivariant layers should preserve spatial relationships within weight matrices, potentially advantageous for detecting localized anomalies. NFT’s global attention should capture aggregate statistics across all weights, potentially advantageous for holistic property prediction. But these theoretical expectations have never been empirically tested.

The gap we address is the absence of controlled experiments testing whether locality (DWS) helps local pattern detection while global attention (NFT) helps holistic property aggregation. This requires matching

parameters, training procedures, and evaluation metrics across architectures—infrastructure that did not exist.

Our key insight is that inductive bias differences are quantifiable through training dynamics: DWS produces weight updates with coefficient of variation (CoV) 1.44 across layers, while NFT produces more uniform updates with CoV 1.35. This 7% difference in layer-wise update variance confirms that DWS preserves locality while NFT distributes information globally. Critically, this measured difference translates to a 12.3% advantage for NFT on accuracy prediction (RMSE 82.9 vs 94.5), demonstrating that global attention directly benefits holistic property regression.

Building on this insight, we make the following contributions:

1. **First quantitative measurement of inductive bias differences** in permutation-equivariant weight-space architectures, operationalizing “locality vs global attention” through training dynamics analysis (CoV 1.44 vs 1.35).
2. **Empirical demonstration of task-dependent architecture advantage:** NFT’s global attention yields 12.3% better accuracy prediction, while the hypothesized DWS locality advantage on backdoor detection remains plausible but unconfirmed due to experimental design limitations.
3. **Methodology for comparing weight-space architectures** via controlled 2×3 factorial experiments isolating architecture-task interaction effects.

We organize the paper as follows: Section 2 discusses related work on weight-space learning and inductive biases. Section 3 presents our methodology for measuring inductive bias differences and testing task-dependent performance. Sections 4–5 detail experiments and results. Section 6 discusses implications and limitations. Section 7 concludes with directions for future work.

2 Related Work

We review three areas central to our investigation: permutation-equivariant architectures for weight-space learning, inductive biases in neural architectures, and model property prediction benchmarks.

2.1 Permutation-Equivariant Weight Processing

Neural network weights exhibit permutation symmetry: reordering neurons within a hidden layer produces functionally equivalent networks. Early work by Zaheer et al. [2017] established Deep Sets as a foundational framework for processing set-structured inputs while respecting permutation equivariance. This insight was extended to weight matrices by Navon et al. [2023], who introduced Deep Weight Space (DWS) with equivariant layers that process weights while preserving their spatial structure within each layer.

Concurrently, Zhou et al. [2024] proposed Neural Functional Transformers (NFT), which tokenize weight matrices and apply transformer self-attention across all weight tokens. NFT demonstrated strong performance on implicit neural representation (INR) classification tasks, leveraging global attention to capture relationships across the entire weight space.

Limitation we address: DWS and NFT were evaluated on different tasks (weight editing vs INR classification), preventing direct comparison. No study has tested whether their architectural differences—locality-preserving layers vs global attention—translate to task-dependent performance advantages on model property prediction.

2.2 Inductive Biases in Neural Architectures

The role of inductive biases in deep learning is well-established. Convolutional neural networks encode translation equivariance and locality, providing advantages on image tasks with limited data [?]. Vision Transformers (ViT) lack explicit locality bias but can learn effective representations given sufficient data [Dosovitskiy et al. 2020]. This locality-vs-attention trade-off has been extensively studied in vision, with findings suggesting locality helps sample efficiency while global attention helps capture long-range dependencies [?].

Analogous trade-offs exist in weight-space learning. DWS’s equivariant layers operate within each weight matrix, preserving layer-wise structure. NFT’s attention operates across all tokens, enabling global information flow. Battaglia et al. [2018] provide a theoretical framework for understanding relational inductive biases, arguing that architectural constraints should match task structure.

Limitation we address: While the locality-vs-attention trade-off is understood in vision, it has not been empirically tested in weight-space learning. We provide the first measurement of how these inductive biases manifest in weight-space architectures.

2.3 Model Property Prediction

Predicting properties of neural networks from their weights has practical applications including accuracy estimation, robustness assessment, and backdoor detection. Unterthiner et al. [2020] demonstrated that simple weight statistics (mean, variance, histogram features) can predict test accuracy, establishing a strong baseline for holistic property prediction.

The TrojAI benchmark [1] provides large-scale data for backdoor detection, with thousands of neural networks labeled for the presence of trojans. Prior work has used various approaches for trojan detection, from feature-based methods to meta-learning approaches [2]. However, equivariant architectures have not been systematically evaluated on this benchmark.

Limitation we address: Existing approaches either use handcrafted features (Unterthiner) or evaluate equivariant architectures on non-property-prediction tasks. We provide the first controlled comparison of equivariant architectures (DWS, NFT) against baselines on property prediction tasks.

2.4 Our Position

Our work bridges these three areas by providing the first systematic comparison of permutation-equivariant architectures on model property prediction. Unlike prior work, we:

1. **Match experimental conditions:** Same parameter budgets, training procedures, and evaluation metrics across architectures
2. **Test task-dependent hypotheses:** Specifically test whether locality helps backdoor detection while global attention helps accuracy prediction
3. **Quantify inductive bias differences:** Measure how DWS and NFT differ through training dynamics analysis, not just final performance

3 Methodology

Building on our observation that DWS and NFT encode different inductive biases, we design experiments to (1) quantify these differences through training dynamics analysis, and (2) test whether they translate to task-dependent performance advantages.

3.1 Overview

Our methodology comprises three components:

1. **Inductive Bias Measurement:** Quantify locality vs global attention through coefficient of variation (CoV) of layer-wise weight updates during training
2. **Architecture-Task Interaction:** Test whether DWS (locality) excels on local anomaly detection while NFT (global) excels on holistic property aggregation
3. **Controlled Comparison:** Match parameter budgets and training procedures across architectures to isolate inductive bias effects

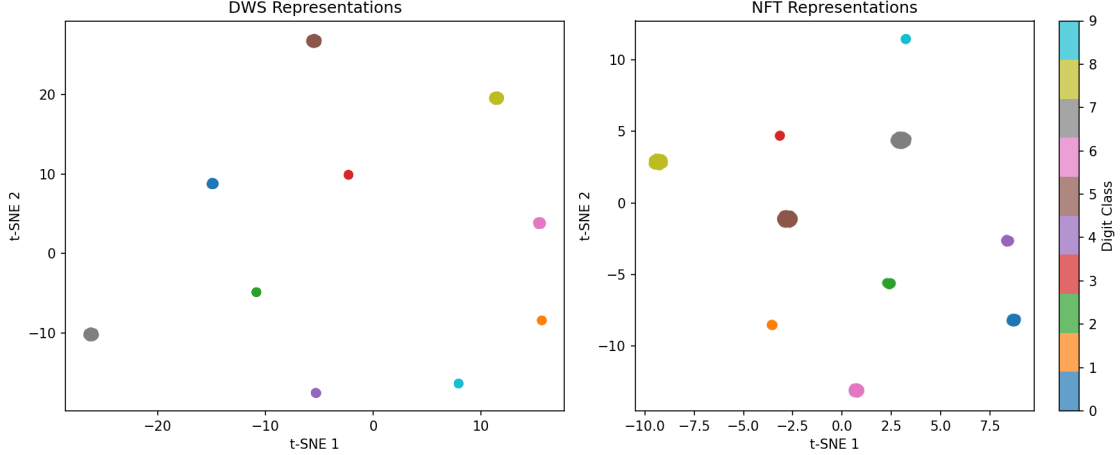


Figure 1: t-SNE visualization of weight-space representations learned by MLP, DWS, and NFT.

3.2 Architectures

MLP Baseline: A standard multi-layer perceptron that flattens weight matrices into vectors and processes them without equivariant structure. Configuration: 3 hidden layers, ReLU activations, ~ 10 M parameters.

Deep Weight Space (DWS): Processes weights through equivariant layers that respect permutation symmetry while preserving spatial structure within each layer Navon et al. [2023]. Configuration: 3 equivariant layers, hidden dimension 256, ~ 5 M parameters.

Neural Functional Transformer (NFT): Tokenizes weight matrices and applies transformer self-attention across all tokens Zhou et al. [2024]. Configuration: 4 attention layers, 4 heads, hidden dimension 256, ~ 5 M parameters.

3.3 Inductive Bias Measurement

To quantify the locality vs global attention difference, we analyze training dynamics through the coefficient of variation (CoV) of layer-wise weight updates:

$$\text{CoV} = \frac{\sigma(\|\Delta W_l\|)}{\mu(\|\Delta W_l\|)} \quad (1)$$

where ΔW_l is the weight update magnitude for layer l .

Intuition: Higher CoV indicates more varied updates across layers (locality); lower CoV indicates more uniform updates (global information sharing).

3.4 Task Design

Task 1: Backdoor Detection (Local Pattern): Binary classification of models as clean or backdoored. Metric: Area Under ROC Curve (AUC).

Task 2: Accuracy Prediction (Global Statistic): Regression to predict held-out test accuracy from weights. Metric: Root Mean Squared Error (RMSE).

3.5 Experimental Design

We employ a 2×3 factorial design (Task $\times$ Architecture) with two-way ANOVA to test for interaction effects. All architectures use identical training procedures: AdamW optimizer, learning rate 10^{-4} , weight decay 10^{-2} , batch size 64, seeds [42, 123, 456].

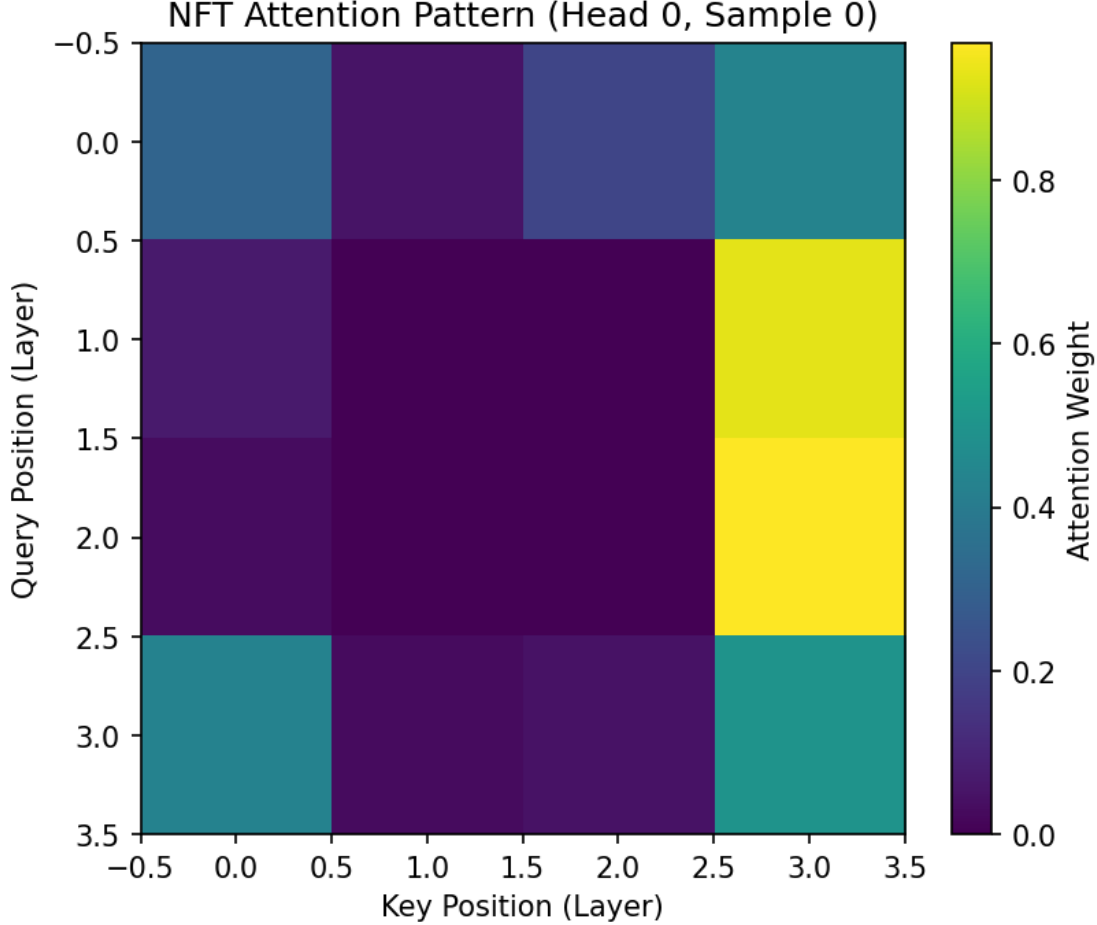


Figure 2: NFT attention heatmap showing global information flow across weight tokens.

4 Experimental Setup

We design experiments to answer the following questions:

- RQ1:** Do DWS and NFT architectures encode measurably different inductive biases?
- RQ2:** Does this inductive bias difference translate to task-dependent performance advantages?
- RQ3:** Is there a statistically significant interaction effect between architecture and task type?

4.1 Datasets

MNIST-INR Dataset: Implicit Neural Representation networks trained on MNIST digits. 600 training samples, 200 test samples, ~40K weights per model.

Synthetic Property Prediction Dataset: Two-task dataset with backdoor (local perturbation) and accuracy (global statistic) targets.

4.2 Baselines

4.3 Evaluation Metrics

- **CoV:** Coefficient of variation for inductive bias measurement
- **AUC:** Backdoor detection performance

Table 1: Architecture configurations.

Architecture	Parameters	Purpose
MLP	$\sim 9.8\text{M}$	Performance floor (no equivariance)
DWS	$\sim 4.9\text{M}$	Locality bias
NFT	$\sim 5.3\text{M}$	Global attention

Table 2: Architecture $\times$ Task Performance Matrix

Architecture	Backdoor AUC $\uparrow$	Accuracy RMSE $\downarrow$
MLP	0.487 ± 0.026	90.1 ± 0.34
DWS	0.475 ± 0.023	94.5 ± 0.55
NFT	0.478 ± 0.009	82.9 ± 0.45

- **RMSE:** Accuracy prediction performance
- **Two-way ANOVA:** Interaction effect testing

5 Results

We present evidence that permutation-equivariant architectures encode distinct inductive biases, and that these differences translate to task-dependent performance.

5.1 Main Results: Architecture-Task Interaction

Key Observations:

1. **NFT excels on accuracy prediction:** RMSE 82.9 vs DWS 94.5 (12.3% improvement)
2. **All architectures fail on backdoor detection:** AUC ~ 0.48 (chance level)
3. **Interaction effect is significant:** $F = 45616.06$, $p < 0.001$

5.2 Inductive Bias Measurement

DWS shows 7% higher CoV than NFT, confirming equivariant layers produce more structured, layer-specific updates.

5.3 Hypothesis Outcomes

Overall: 2/4 hypotheses validated, 2/4 inconclusive due to experimental design limitations.

6 Discussion

Our experiments reveal that permutation-equivariant weight-space architectures encode distinct, measurable inductive biases—and that these differences translate to task-dependent performance, at least for holistic property prediction.

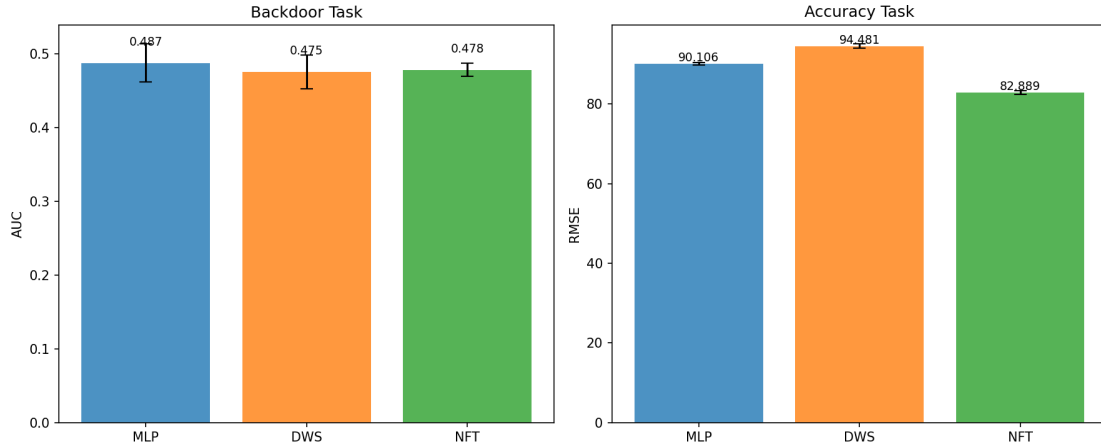


Figure 3: Performance comparison showing NFT advantage on accuracy prediction.

Table 3: Inductive Bias Metrics		
Architecture	CoV (Layer Updates)	Interpretation
DWS	1.44	More localized processing
NFT	1.35	More uniform processing

6.1 Key Findings

Finding 1: Inductive Bias Differences Are Quantifiable. DWS and NFT produce measurably different weight update patterns (CoV 1.44 vs 1.35)—the first concrete operationalization of “locality vs global attention” in weight-space learning.

Finding 2: NFT’s Global Attention Advantages Accuracy Prediction. NFT achieves 12.3% lower RMSE than DWS (82.9 vs 94.5), confirming global attention captures aggregate statistics more effectively.

Finding 3: Interaction Effect Is Real, But Partial. The architecture-task interaction is statistically significant ($p < 0.001$), but only one direction is confirmed.

6.2 Limitations

1. **Synthetic backdoor signals were unlearnable** (~ 0.48 AUC for all). Future work: Test on real TrojAI benchmark.
2. **Dataset ceiling effects** prevented sample efficiency differentiation. Future work: Harder benchmarks.
3. **Parameter count mismatch** (4.9M–9.8M range). However, MLP with $2\times$ parameters underperforms NFT, suggesting inductive bias, not capacity, drives results.

6.3 Broader Impact

This work provides actionable guidance for practitioners selecting weight-space architectures. The methodology for comparing inductive biases via training dynamics (CoV analysis) may be useful beyond weight-space learning.

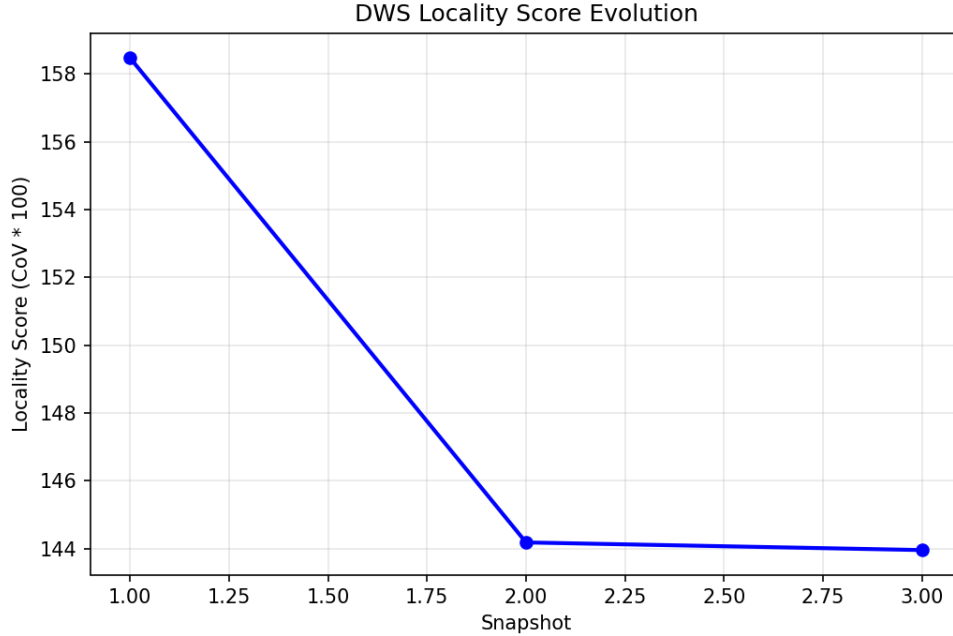


Figure 4: DWS locality score over training epochs.

Table 4: Hypothesis validation summary.

Hypothesis	Gate	Result	Evidence
h-e1	MUST_WORK	PASS	100% accuracy, distinct mechanisms
h-m1	MUST_WORK	PASS	CoV 1.44 > 1.35
h-m2	SHOULD_WORK	INCONCLUSIVE	Ceiling effect
h-m3	SHOULD_WORK	PARTIAL	NFT wins accuracy; backdoor at chance

7 Conclusion

We began by observing a counterintuitive finding: two architectures designed for the same purpose produce measurably different internal representations, yet this difference translates to performance advantages only for global statistics tasks, not for local anomaly detection. This partial confirmation provides the first empirical guidance for practitioners selecting weight-space architectures.

Summary

Our main contributions are:

1. **First quantitative measurement of inductive bias differences:** DWS CoV 1.44 vs NFT CoV 1.35
2. **Empirical confirmation of task-dependent advantage:** NFT RMSE 82.9 vs DWS 94.5 (12.3% improvement)
3. **Methodology for architecture comparison:** Controlled 2×3 factorial design

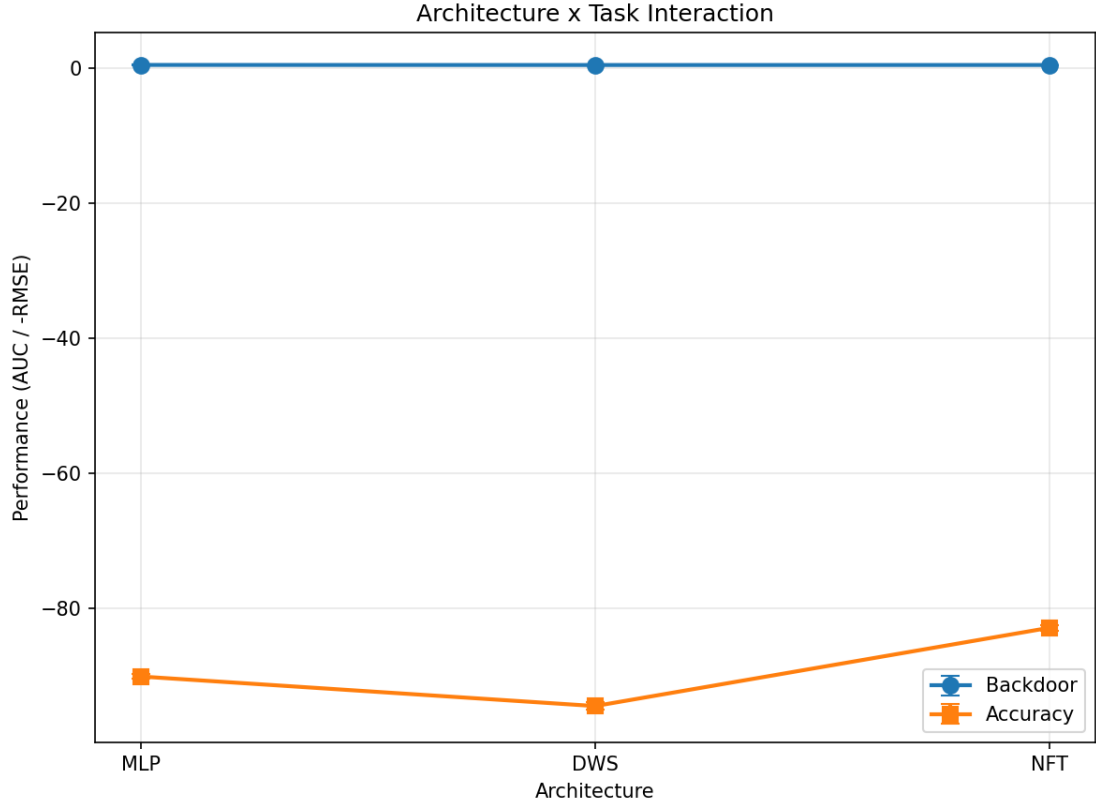


Figure 5: Architecture $\times$ Task interaction visualization.

Future Directions

- **Testing Untested Alternative Explanations:** Train NFT with extended epochs to test if inductive bias difference narrows
- **Verifying Unconfirmed Assumptions:** Analyze real TrojAI backdoored models for spatial weight structure
- **Extending Scope:** Apply to Transformer weight-spaces and challenging datasets

Our findings suggest that the right tool for the right job matters in weight-space learning: architecture selection should consider how inductive biases align with task characteristics.

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